Image Processing Pipelines for Offline and Online 3D Reconstruction with a Monocular USB Camera

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Outline

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- Offline Calibration (used for both pipeline)
- Online Image Processing Pipeline
- Offline Image Processing Pipeline
- Experiments
- Demo
- Future work

Abstract

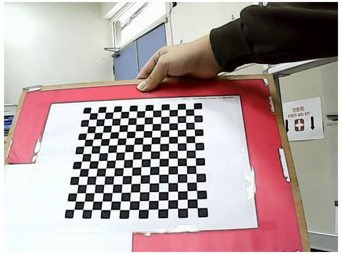
- Goal: Improve 3D reconstruction stability and accuracy using a USB camera in both online (direct visual SLAM) and offline (COLMAP) settings.

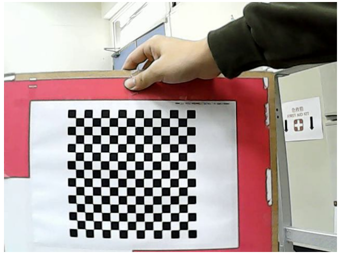
- Online: Improve the stability of 3D reconstruction tailored for direct visual SLAM.
=> Enhance photometric and geometric consistency without sacrificing real-time performance.

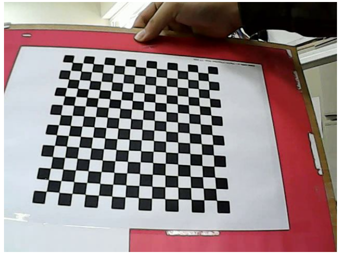
- Offline: Improve the quality of 3D reconstruction for COLMAP feature-based mechanism,
=> Prioritize feature density over the perceptual quality.

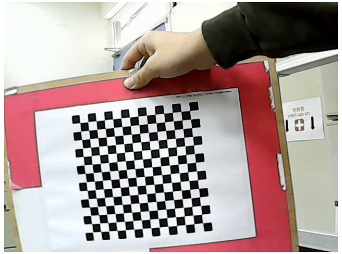
Camera Calibration

- Method: Zhang’s calibration method
- A checkerboard calibration target with 14 × 14 inner corners, corresponding to squares of size 12 mm.
- Number of input images: Hundreds
- fx: 657.7274, fy: 658.6482, cx: 339.5179, cy: 239.1852, and distortion coefficients





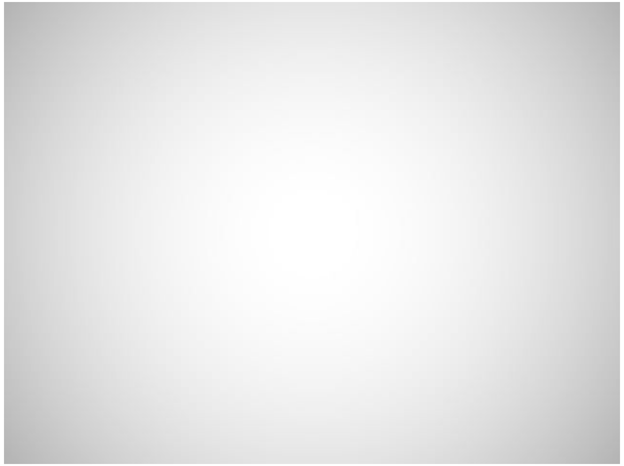




Photometric Calibration

- The method can process auto exposure videos with visual odometry precisions.

- It leverages a image formation model to link intensities to photometric parameters (per-frame exposure, vignetting, ...) followed by nonlinear optimizations



Why photometric caibration?

To eliminate position-dependent brightness decay.
∵ It creates false photometric errors.

Online Photometric Calibration of Auto Exposure Video for Realtime Visual Odometry and SLAM (P. Bergmann, R. Wang, D. Cremers), In IEEE Robotics and Automation Letters (RA-L), volume 3, 2018.

What is DSO? (Online)

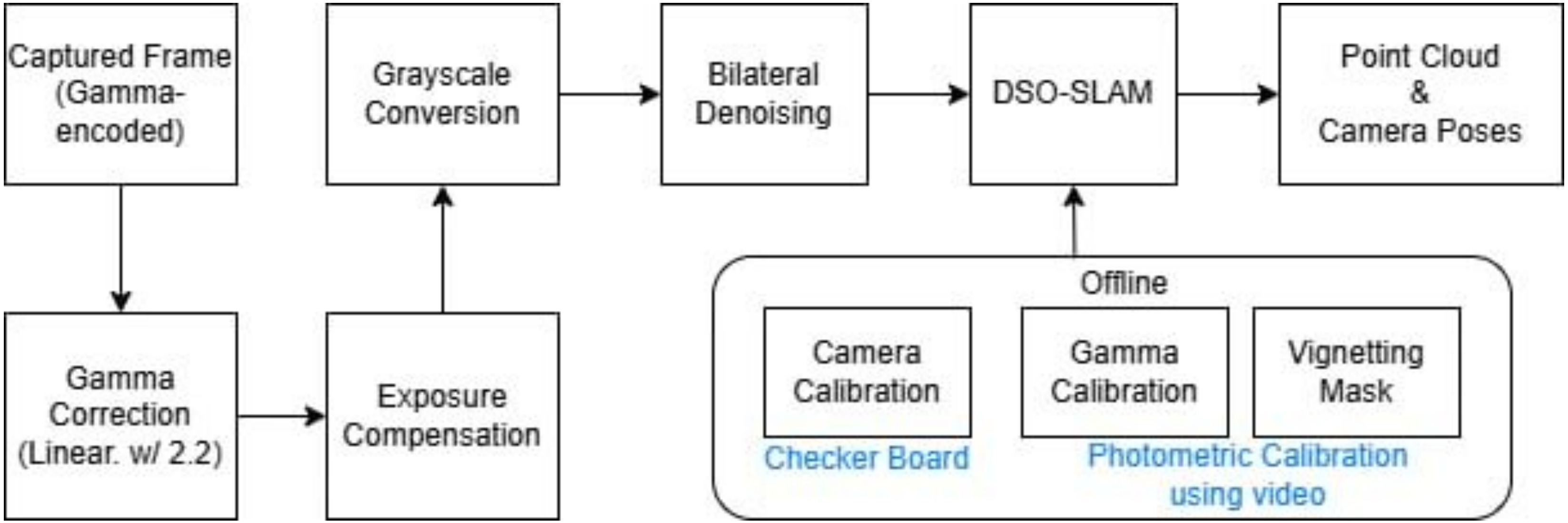
- DSO estimates camera motion by directly aligning pixel intensities.
- It minimizes photometric error on a sparse set of high-gradient pixels.
- No feature detection is involved.
- The core assumption :
  the same 3D point has consistent brightness across frames.

Why image processing pipeline matters?

- Frames must be photometrically consistent to make intensity alignment meaningful.
- Nonlinear image processing can break this consistency to some extent.
- Geometric and photometric corrections are essential for stable optimization.

Online Pipeline

Overview :



Differences Between Normal Image Process Pipeline

- General ISP: Produces images that are visually pleasing for humans, with natural colors and high subjective visual quality.

- Our image processing pipeline: Designed specifically to serve DSO (Direct Sparse Odometry)

- White balance: Implemented as per-channel gain, which introduces unnecessary nonlinearity (✗)

- Edge enhancement / sharpening: Nonlinear amplification of high-frequency components, directly violating photometric gradient consistency (✗)

Differences Between Normal Image Process Pipeline

- Exposure Compensation: Introduces a temporal exposure stabilization mechanism to ensure that adjacent frames are comparable on the same brightness scale.
- Vignetting Correction: Explicitly removes the systematic bias where the same 3D point appears with different brightness due to its image location.

Processing Step	Conventional ISP	Proposed pipeline	Reason
Auto White Balance	✓	✗	Unnecessary nonlinearity transformation.
Color Interpolation / Correction	✓	✗	DSO doesn’t require full-color reconstruction
Gamma correction	✓	✓	Increase robustness
Saturation Enhancement	✓	✗	Modifies intensity distribution
Tone Mapping	✓	✗	Scene-dependent nonlinearity transformation
Edge Enhancement / Sharpening	✓	✗	Artificially amplifies gradients
Bilateral Denoising	✗	✓	Reduces noise while preserving true image gradients
Exposure Compensation	✗	✓	Aligns brightness scale between consecutive frames
Vignetting Correction	✗	✓	Removes position-dependent brightness bias
Lens Undistortion	✗	✓	Ensures correct pixel correspondence for direct alignment.

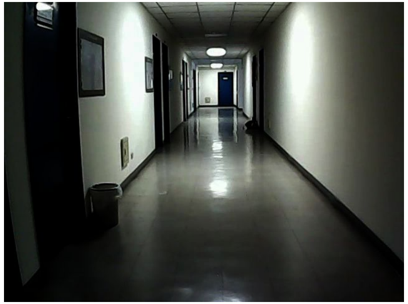
Gamma Correction

- A fixed intensity mapping is applied to the input pixels to make the distribution of dark and bright regions more controllable.

- We use a fixed gamma value of 2.2 as a standard approximation for sRGB cameras

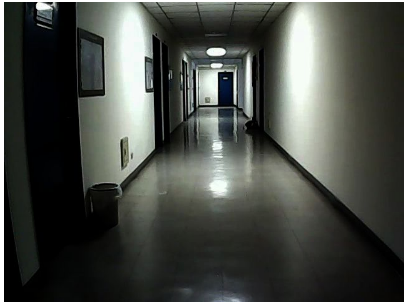
- Gamma correction here serves as the tool to map gamma-encoded data to linear radiance values, ensuring accurate photometric modeling for DSO's direct method

Before

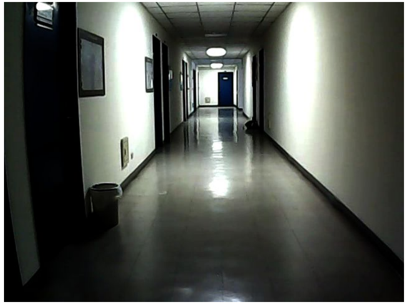


After

Exposure Compensation



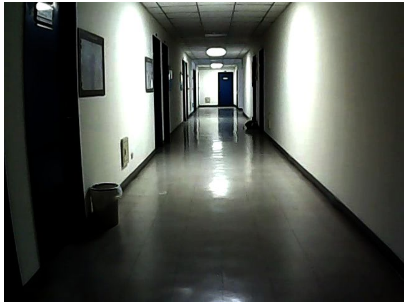
Before



After

- A global gain is applied to stabilize frame-to-frame brightness variations while maintaining real-time performance.
- The gain is computed using an exponential moving average (EMA) based on median luminance instead of mean luminance.
- An adaptive update rate determined by an instability measure computed from a 3x3 grid of image regions is used.

Grayscale Convertion & Bilateral Denoising



Before

After
Grayscale Conversion

After
Bilateral Denoising

- Grayscale conversion is performed using the BT.709 luminance formulation.
- A Bilateral filter is applied for light smoothing.
- Light Bilateral smoothing suppresses sensor noise and stabilizes gradient estimation without removing the high-gradient structures that are critical for direct alignment.

Undistortion

- Undistortion ensures that a pixel observed in frame t is mapped to the correct corresponding location in frame t+1.

- Since DSO computes intensity differences between corresponding pixels, incorrect correspondence locations directly increase photometric error and degrade reconstruction performance.

Before

After

Offline Version

- The performance of COLMAP may degrade in the presence of distortion and sensor noise.
- Existing image processing methods are pursuing perceptual quality, which surpass noise and contrast. The method will have negative impact on COLMAP, which relies on gradients for feature extractions.
- To tackle with the problem, we integrate CLALHE to enhance images before COLMAP process.

Differences Between Normal Image Process Pipeline

- Normal image preprocessing prioritizes human perceptual quality but may not be suitable for feature extraction.

- We introduce an image processing pipeline tailored to improve feature extraction and matching for feature-based 3D reconstruction.

- Naïve Gaussian blurring for denoising is counterproductive, as it suppresses edge information and reduces keypoint localization accuracy.

Bilateral Filter

- Raw images often contain high-frequency noise, which can be misidentified by COLMAP as feature points, leading to false-positive keypoints.
- The bilateral filter smooths low-texture regions while preserving edge structures, thereby reducing unnecessary computation on noise-induced features.
- Since SIFT relies on gradient histograms, the use of bilateral filtering ensures sharp edge preservation, resulting in more stable keypoint orientation estimation.

CLAHE
(Contrast Limited Adaptive Histogram Equalization)

- CLAHE method improves features by Boosting Local Texture
- Also, it enhances local contrast, which makes enhance COLMAP feature matching

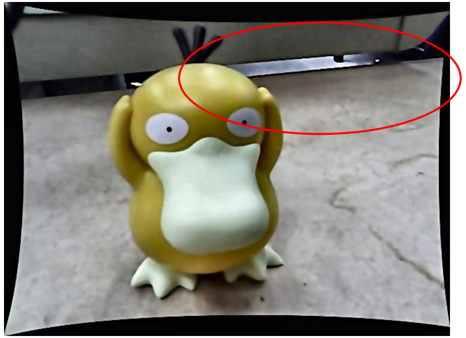
Raw

After CLAHE

Advanced CLAHE: CLALHE

- CLAHE employs fixed parameters (e.g., ClipLimit = 2.0) without considering image content.
  - In flat regions, this leads to excessive amplification of sensor noise, producing false features.
  - In complex dark regions, the enhancement strength is insufficient, and valid feature points remain undetected.

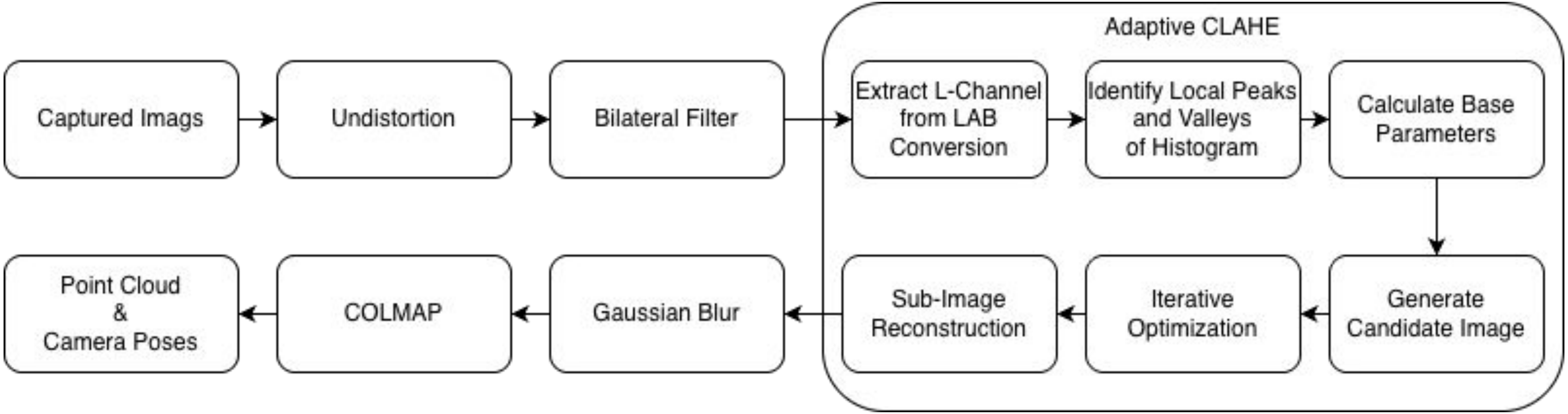
- Hence, a new approach called CLALHE (L stands for Local) was introduced to tackle with the problems of CLAHE.



Gaussian Blur

- The Gaussian blur applied here is used specifically for **unsharp masking**.
- The extracted edge components are added back to the original image with a fixed strength, thereby mathematically increasing the contrast of all edges.
- This enhances feature robustness and track length, both of which are critical for constructing accurate 3D models.

Offline Pipeline (CLALHE)



Differences Between CLALHE and CLAHE

	Standard CLAHE	Proposed CLALHE
Optimization Goal	Perceptual Naturalness	Geometric Utility
Contrast Intensity	Conservative	Aggressive / Boosted
Sub-Image Reconstruction	Seamless Blending	Independent Concatenation

CLALHE: Comparison

Raw

CLAHE

CLALHE

Experiments (Online)

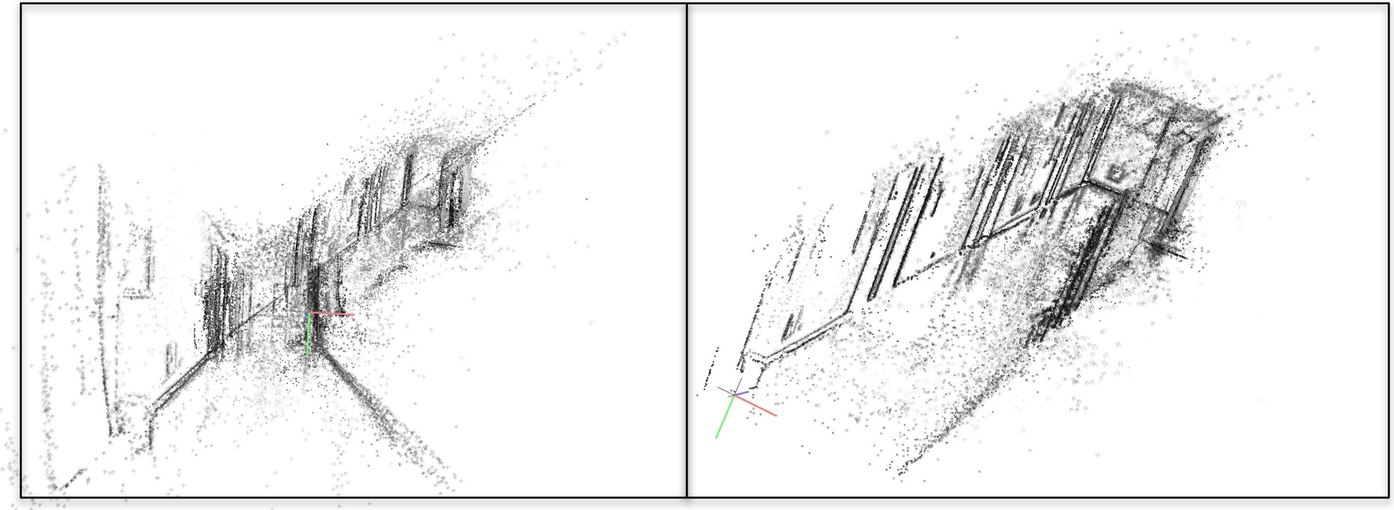
- Online system operates at the camera’s native frame rate (~30fps), with all processing executed in real-time on the host CPU.

- Scene : Long corridor, which is one of selected scene in DSO paper.

Experiments (Online)

Raw frame

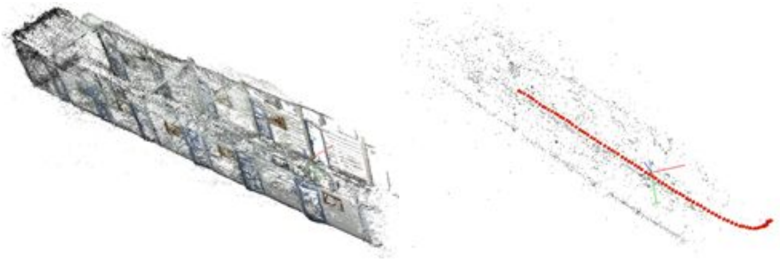
Our method



Experiments (Offline)

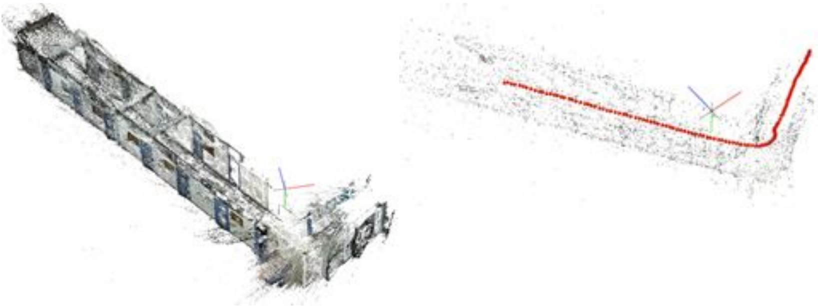
Method	Reproj. Error	Reg. Rate	# Points (Sparse / Dense)	Mean Obs. / Image
Raw	**0.944**	41.8% (84)	4.1k / 0.18M	**273.7**
Our (CLALHE)	1.011	41.8% (84)	4.3k / 0.15M	263.1
Our (CLAHE)	1.006	**63.2% (127)**	**5.6k / 0.26M**	247.0

Raw frame



Our method (CLALHE)

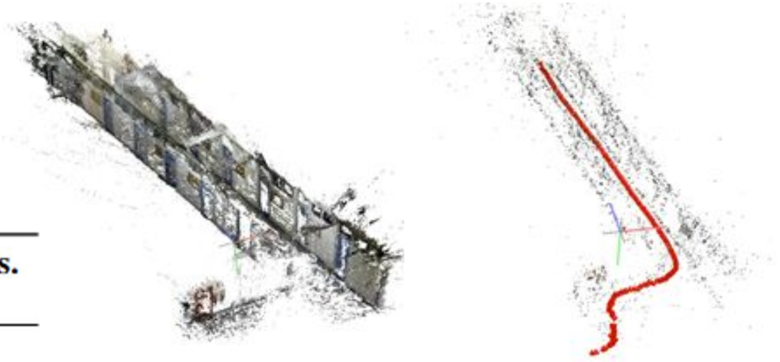
Our method (CLAHE)



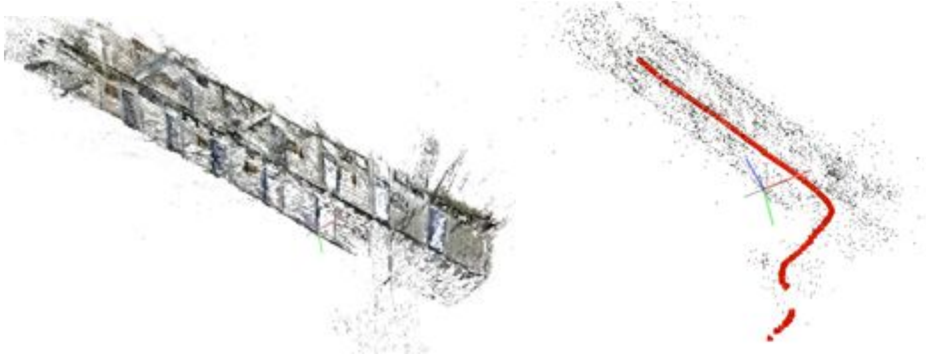
Experiments (Offline)

Method	Reproj. Error	Reg. Rate	# Points (Sparse / Dense)	Mean Obs. / Image
Raw	**0.964**	28.1% (148)	6.0k / 0.27M	236.9
Our (CLALHE)	1.045	26.3% (138)	6.5k / 0.23M	**246.8**
Our (CLAHE)	1.043	**47.0% (246)**	**8.0k / 0.40M**	187.1

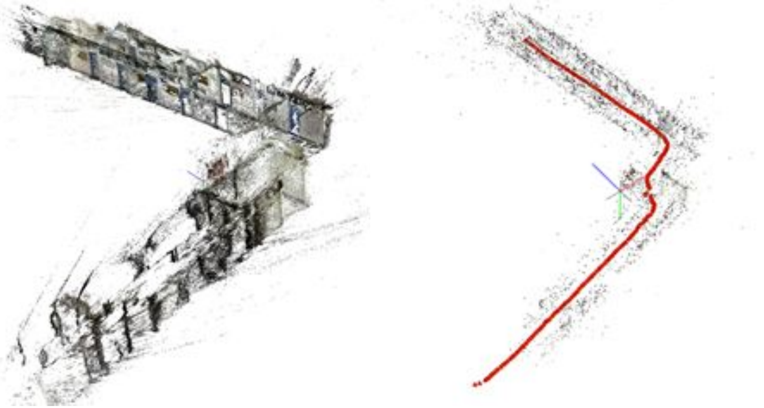
Raw frame



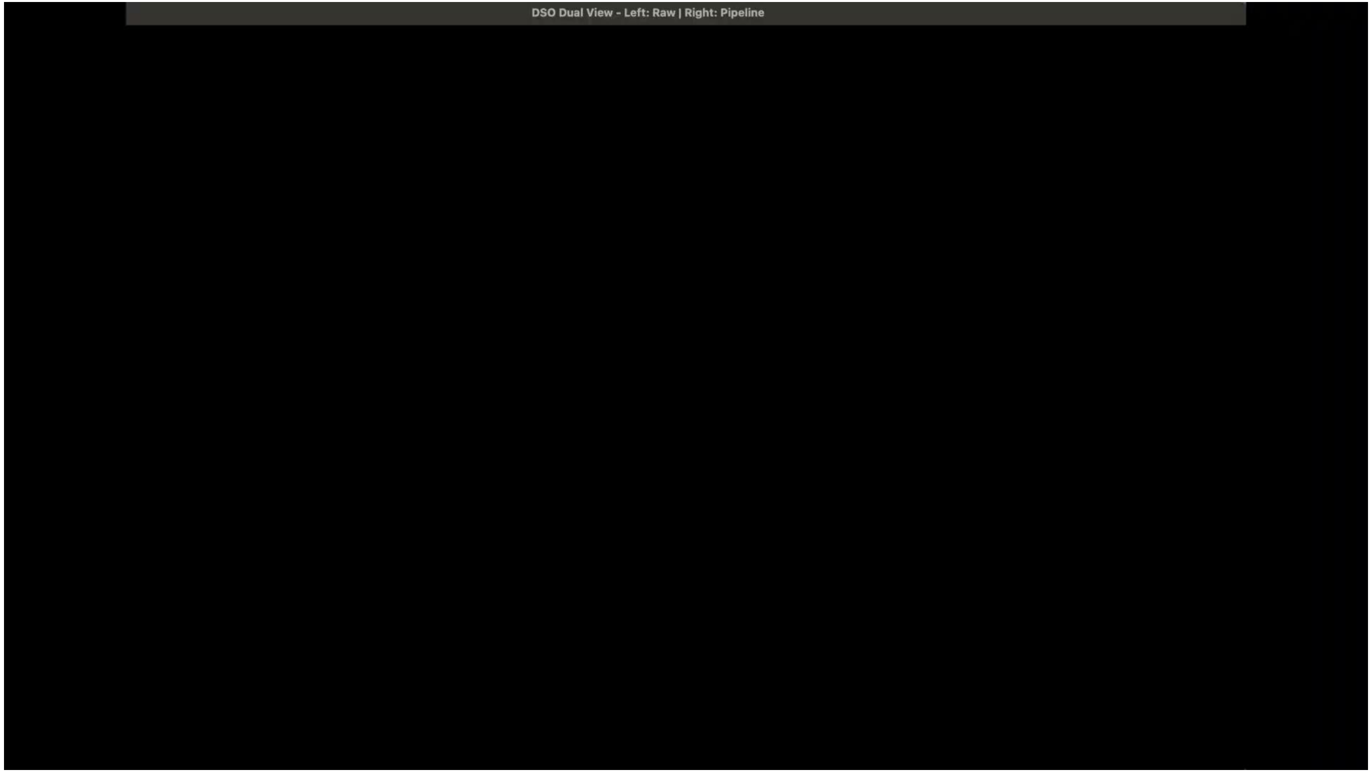
Our method (CLALHE)



Our method (CLAHE)



Demo (Online)



Discussion (Online)

1. Point cloud divergence at initialization
    1.1. Indicates unstable photometric alignment during the initialization phase

2. Unstable camera trajectory
    2.1. The estimated camera pose shows significant jitter and drift at the beginning of the sequence.

3. Scale shift in later frames

Discussion (Offline)

Although the feature point per frame have improved from simple CLAHE, here came several flaws:

1. The grid shown in tuned CLALHE might have severely impacted the consistency of image when undergoing COLMAP 3D reconstruction, making the path terminates prematurely

2. The proposed method failed to maintain the continuity of the track in the outdoor / hallway scenario.

3. Sharp vertical lines violate SfM assumptions despite much more features were discovered

Future work

1. Online : The system still fail under extreme lighting conditions such as motion-activated lights. Future work will focus on improving robustness under abrupt and non-smooth illumination changes, potentially by incorporating more adaptive exposure modeling.

2. Offline: To address the issues of path breakage and blocking artifacts (the grid lines) while maintaining the high feature count per image, we could try :

   Make the contrast not so aggressive and eliminates the sharp vertical gradients
   => preventing COLMAP from detecting the grid lines as static 3D features

References

[1]. Jakob Engel, Vladlen Koltun, and Daniel Cremers, “Direct sparse odometry,” IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 40, no. 3, pp. 611–625, 2018.

[2]. Bergmann, R. Wang, and D. Cremers, “Online photometric calibration of auto exposure video for realtime visual odometry and slam,” IEEE Robotics and Automation Letters (RA-L), vol. 3, pp. 627–634, 2018.

[3]. Garima Yadav, Saurabh Maheshwari, and Anjali Agarwal, “Contrast limited adaptive histogram equalization based enhancement for real time video system,” in 2014 International Conference on Advances in Computing, Communications and Informatics (ICACCI), 2014, pp. 2392–2397.

[4]. Johannes L. Sch¨onberger and Jan-Michael Frahm, “Structure-from-motion revisited,” in 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016, pp. 4104–4113.